ORBITAL-X FLIGHT OPERATIONS // MANUAL REF: OX-TC-2026-V4

Satellite Telecommand & Flight Communications Reference Manual

Complete Command Vocabulary, Telemetry Packet Formats, Ground Station Uplink Protocols & Emergency Recovery Procedures

This document provides the definitive operational specifications for communicating with CubeSats and orbital assets managed under the Orbital-X Mission Operations System. It details every valid telecommand opcode, argument schemas, expected physical and software telemetry effects, radio frequency bands, CCSDS packet framing, and emergency contingency procedures.

APPLICABLE ASSETS

ApexSat-1 (SAT-ORB-01), ApexSat-2 (SAT-ORB-02), Generic 3U/6U CubeSats

FLIGHT SOFTWARE VERSION

OrbitalX-FSW v3.8.4 (SGP4/Keplerian Solvers Integrated)

COMMUNICATION STANDARDS

CCSDS 231.0-B-3 (TC Framing), AX.25, UHF/S/X-Band Duplex

CLEARANCE LEVEL

Mission Operator Clearance / Flight Director Execution Authority

CLASSIFICATION: OPERATIONAL USE ONLY ORBITAL-X MISSION CONTROL CENTER OCTOBER 2026

1. Telecommand Architecture & Uplink Pipeline

SECTION 1.0

Satellite telecommands in Orbital-X originate from the Mission Control Console or automated scripts, pass through security authentication and permission verifications, and are dispatched via REST or WebSocket to ground station software-defined transceivers for RF modulation.

```
TELECOMMAND DISPATCH FLOW

[MCC Operator / Script] ──(HTTPS Bearer Token)──> [OrbitalX API Gateway :4000]
  └── Auth Verification & Clearance Check (Requires Role: OPERATOR / ADMIN)
  └── Endpoint: POST /api/telemetry/cubesat/command
      ├── [Audit Logger] Records timestamp, operator ID, satellite ID, opcode
      ├── [Ground Station Router] Evaluates pass window, elevation > 10°, and Doppler offset
      └── [Onboard Telemetry Simulator] Evaluates physical effects, battery, ADCS attitude, thrust & temp
```

1.1 REST Telecommand Ingest Interface

Operators and automated flight systems execute telecommands via standard authenticated HTTP payloads:

```
// Dispatched to: POST /api/telemetry/cubesat/command // Headers: Authorization: Bearer <TOKEN> | Content-Type: application/json { "satelliteId": "SAT-ORB-01", "command": "COLD_GAS_THRUST", "params": { "deltaV": 0.08, "durationSec": 3 } }
```

1.2 Radio Frequency & Ground Station Link Parameters

LINK TYPE	FREQUENCY BAND	DOWNLINK FREQ	UPLINK FREQ	MODULATION / PROTOCOL	MAX DATA RATE
Telemetry Beacon	UHF (Amateur/CubeSat)	437.525 MHz	N/A (Broadcast)	AFSK / AX.25 (G3RUH)	9.6 kbps
Telecommand Uplink	UHF / S-Band	N/A	435.150 MHz / 2025 MHz	GFSK / CCSDS Packet	38.4 kbps
Payload Data Downlink	X-Band	8450.000 MHz	N/A	OQPSK / DVB-S2	150.0 Mbps
Ground Stations	Bengaluru (GS-BLR-01: 12.97°N, 77.59°E) • Svalbard (GS-SVB-02: 78.22°N, 15.65°E) • Houston (GS-HOU-03: 29.76°N, -95.36°E)				

> **DOPPLER SHIFT COMPENSATION REQUIREMENT**
> For Low Earth Orbit (LEO) passes at 500 km altitude (orbital velocity ~7.6 km/s), ground station transmitters must apply real-time frequency correction of up to **±9.5 kHz on UHF** and **±45 kHz on S-Band** throughout the 10-minute pass window to prevent receiver PLL lock dropouts.

2. Comprehensive Satellite Telecommand Vocabulary

SECTION 2.0

The following tables detail all valid commands parsed and executed by the Orbital-X Onboard Computer (OBC) and physics simulation engine. Commands may be issued using the canonical opcode or standard alias shortcuts.

2.1 Electrical Power Subsystem (EPS) Commands

COMMAND OPCODE	VALID ALIASES	PARAMETERS	ONBOARD PHYSICAL EFFECT & TELEMETRY RESPONSE	CLEARANCE
DEPLOY_SOLAR_PANELS	DEPLOY_PANELS, DEPLOY	None	Sets `solarPanelsDeployed = true`. Increases solar generation (+35W). Solar panel temperature rises nominal (+3.2°C). Battery charging accelerates.	OPERATOR
RETRACT_SOLAR_PANELS	STOW_PANELS, STOW	None	Sets `solarPanelsDeployed = false`. Stows array into chassis for protective high-drag passes. Reduces bus power input.	FLT DIR
CMD_EPS_SAFE_MODE	EPS_SAFE, POWER_CONSERVE	None	Sheds non-essential sensor rails. Restricts power draw strictly to OBC and UHF receiver (< 6.4W).	FLT DIR
CMD_EPS_SWITCH_BUS_B	BUS_B_TOGGLE	None	Switches regulated power distribution to redundant secondary 5V/12V converter rails.	ADMIN

2.2 Attitude Determination & Control System (ADCS) Commands

COMMAND OPCODE	VALID ALIASES	PARAMETERS	ONBOARD PHYSICAL EFFECT & TELEMETRY RESPONSE	CLEARANCE
ADCS_DETUMBLE	DETUMBLE	None	Activates B-dot magnetic torquer coils. Nulls tumbling spacecraft angular rates below **0.01°/s**. Sets mode to `DETUMBLED`. Spikes CPU load +22% for control loop.	OPERATOR
POINT_NADIR	NADIR	None	Slews satellite body Z-axis directly toward Earth center (Nadir). Sets Roll: 0.1°, Pitch: 0.0°, Yaw: 0.0°. Required for optical imaging passes and high-gain ground antenna downlinks.	OPERATOR
POINT_SUN	SUN, SUN_TRACK	None	Aligns solar panel normal vector to the Sun vector. Optimizes solar energy absorption (+4.8°C solar panel temp).	OPERATOR
CMD_ADCS_RESET_GYRO_BIAS	RESET_GYRO	None	Re-calibrates MEMS gyroscope zero-rate offset using Kalman filter star-tracker attitude truth.	FLT DIR

2.3 Propulsion & Orbital Maneuvers

COMMAND OPCODE	VALID ALIASES	PARAMETERS	ONBOARD PHYSICAL EFFECT & TELEMETRY RESPONSE	CLEARANCE
COLD_GAS_THRUST	ORBITAL_BOOST, THRUST, BURST	`deltaV` (km/s) `durationSec` (s)	Fires micro-thruster nozzle for 3 seconds. Imparts **+0.08 km/s** speed boost. Raises orbital altitude +3.8 km. Consumes 2.4% cold-gas propellant. Status transitions to `THRUSTER_BURN`, then auto-recovers to `NOMINAL`.	FLT DIR / ADMIN
CMD_PROP_ARM_CAM_BURN	ARM_CAM	`burnVector` [x,y,z]	Arms emergency collision avoidance maneuver (CAM) valve sequence. Requires dual operator confirmation.	FLT DIR

PROPELLANT EXPENDITURE CONSTRAINT
Cold-gas propellant is non-renewable. Each firing consumes approximately **2.4% of total tank capacity**. Thruster bursts must only be authorized for orbital altitude station-keeping or certified conjunction avoidance maneuvers.

3. Payload, Communications & System Health Commands

SECTION 3.0

3.1 Scientific Payload & Earth Observation

COMMAND OPCODE	VALID ALIASES	PARAMETERS	ONBOARD PHYSICAL EFFECT & TELEMETRY RESPONSE	CLEARANCE
EXECUTE_PAYLOAD_CAPTURE	CAPTURE_IMAGE, CAPTURE, SCIENCE_PASS	None	Activates multispectral optical imager. Captures Earth surface frames. Spikes CPU load to **~75-96%** and memory usage +26%. Writes 240 MB imagery data to onboard flash buffer.	OPERATOR
START_DATA_DOWNLINK	DOWNLINK, FLUSH_MEMORY	None	Powers X-Band high-gain transmitter. Downlinks **450 science telemetry packets** to active ground station. Reduces onboard memory utilization by 28%. Elevates radio power amplifier temp (+5.2°C).	OPERATOR

3.2 Thermal Control Subsystem (TCS)

COMMAND OPCODE	VALID ALIASES	PARAMETERS	ONBOARD PHYSICAL EFFECT & TELEMETRY RESPONSE	CLEARANCE
CYCLE_HEATERS	HEATERS_ON, HEATERS, TOGGLE_HEATERS	None	Toggles internal resistance heating loops ON/OFF. When ON: Elevates internal bus temperature (+5.8°C) and battery temp (+4.2°C). Draws additional 8.5W bus power. Yellow HUD heater indicator lights up.	OPERATOR
HEATERS_OFF	HEATER_OFF	None	Explicitly de-energizes heating loops. Prevents overheating during high solar irradiance passes.	OPERATOR

3.3 Command & Data Handling (CDH / OBC) & Diagnostics

COMMAND OPCODE	VALID ALIASES	PARAMETERS	ONBOARD PHYSICAL EFFECT & TELEMETRY RESPONSE	CLEARANCE
PING_DIAGNOSTICS	PING	None	Executes full round-trip radio ping & health echo. Returns: `EPS: 100% | CDH: 100% | ADCS: 98% | COMM: 99% | Latency: 14.2ms`.	OPERATOR
ENTER_SAFE_MODE	SAFE_MODE, SAFE	None	Forces spacecraft into safe configuration: powers down science payload, shuts off auxiliary heaters, throttles CPU clock to 8%, sets bus draw to 6.4W.	FLT DIR
RESUME_NORMAL_OPS	NORMAL_OPS, RESUME	None	Clears safe mode flags. Resumes normal 24% CPU baseline and nominal telemetry streaming.	FLT DIR
REBOOT_OBC	REBOOT	None	Performs warm reboot of flight computer processor. Re-initializes RAM, executes firmware checksum verification (`0x8F4A`), and recovers within 3 seconds.	FLT DIR

3.4 Quick Reference Command Syntax Cheat-Sheet

```
// 1. Stabilize Tumbling Satellite POST /api/telemetry/cubesat/command { "satelliteId": "SAT-ORB-01", "command": "DETUMBLE" } // 2. Point Nadir & Start Downlink { "satelliteId": "SAT-ORB-01", "command": "POINT_NADIR" } { "satelliteId": "SAT-ORB-01", "command": "START_DATA_DOWNLINK" }
```

```
// 3. Orbit Raise Altitude Burn POST /api/telemetry/cubesat/command { "satelliteId": "SAT-ORB-01", "command": "COLD_GAS_THRUST" } // 4. Emergency Safe Mode Contingency { "satelliteId": "SAT-ORB-01", "command": "ENTER_SAFE_MODE" }
```

4. Telemetry & Telecommand Packet Specifications

SECTION 4.0

For low-level software-defined radio (SDR) integration, commands are encoded according to the **CCSDS 231.0-B-3 Space Link Protocol**. Each telecommand frame contains a standardized 48-bit primary header followed by the command payload and an error-detecting cyclic redundancy code.

4.1 CCSDS Telecommand Transfer Frame Structure

FIELD NAME	BIT WIDTH	FIELD DESCRIPTION & VALUES
Version Number	3 bits	Set to `000b` for CCSDS Version 1 Transfer Frames
Type Indicator	1 bit	`1b` = Telecommand (Uplink), `0b` = Telemetry (Downlink)
Secondary Header Flag	1 bit	`1b` indicates presence of 32-bit timestamp in secondary header
Application Process ID (APID)	11 bits	`0x010` = EPS Power • `0x020` = ADCS Attitude • `0x030` = Propulsion • `0x040` = Science Payload • `0x050` = OBC CDH
Sequence Flags	2 bits	`11b` = Unsegmented single packet
Packet Sequence Count	14 bits	Monotonically increasing sequence counter (0 to 16383) to prevent replay attacks
Packet Data Length	16 bits	Length of user telecommand payload in bytes minus 1
Command Opcode & Args	Variable	8-bit opcode followed by argument bytes (e.g. `0x1A 0x03` for cold gas thruster burst)
Frame Error Control (CRC-16)	16 bits	CCITT-16 checksum polynomial (`x^16 + x^12 + x^5 + 1`) for frame error rejection

4.2 Ground Station Pass Coordination Checklist

Ground station passes typically last between **8 and 12 minutes** for low-earth satellites. Flight controllers must follow this sequence for every pass:

AOS (Acquisition of Signal) Checklist

1. Antenna dish slews to predicted AOS azimuth / elevation.
2. RF demodulator acquires carrier lock at horizon +10°.
3. Receive and decode broadcast beacon packets (UHF 437.525 MHz).
4. Verify battery state of charge (> 70%) and bus voltage (> 11.4V).
5. Confirm attitude determination mode is `NADIR` or `SUN_TRACKING`.

LOS (Loss of Signal) Checklist

1. Transmit all pending queued telecommands before elevation < 10°.
2. Verify command execution acknowledgments from telemetry stream.
3. Confirm X-Band payload downlink buffer flushed to ground archive.
4. Command payload transmitter OFF to conserve battery reserve.
5. Log pass statistics: total packets, signal SNR, Doppler shift, bit errors.

4.3 Emergency Contingency Matrix

OBSERVED ANOMALY	ROOT CAUSE DIAGNOSIS	IMMEDIATE CORRECTIVE TELECOMMAND SEQUENCE
`TUMBLE RATE > 15°/S`	Reaction wheel saturation or separation tip-off	1. `ENTER_SAFE_MODE` 2. `ADCS_DETUMBLE` (Energizes B-dot coils) 3. Await body rate < 0.1°/s, then issue `POINT_SUN`
`BATTERY < 20%`	Subsystem eclipse drain or array misalignment	1. `HEATERS_OFF` (Kill auxiliary thermal loads) 2. `DEPLOY_SOLAR_PANELS` (Verify full array exposure) 3. `POINT_SUN` (Maximize solar angle of incidence)
`CPU HANG / TELEMETRY FREEZE`	Radiation single-event latchup in flight software	1. Transmit `PING_DIAGNOSTICS` on hardware carrier 2. Issue `REBOOT_OBC` to trigger golden image bootloader 3. Verify checksum response: `0x8F4A` verified